Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/15/2026
PI Name:	Blanchard, Joel W Ph.D.		
Project Title:	Identification of Biomarkers and Pathological Mechanisms via Longitudinal Analysis of Neurological and Cerebrovascular Responses to Neurotoxic Stress Using a Multi-cellular Integrated Model of the Human Brain		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	10029	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	09/28/2022	**End Date:**	09/29/2026
No. of Post Docs:	3	**No. of PhD Degrees:**	4
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
Contact Email:	ursula.m.koniges@nasa.gov		
Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	No Changes		
COI Name (Institution):	Han, Woojin Ph.D. (ICAHN School Of Medicine at Mount Sinai)		
Grant/Contract No.:	80ARC022CA004		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As we age, our brains are naturally exposed to increasing amounts of neurotoxic insults, such as DNA damage, neuroinflammation, and amyloid-beta. In many, the buildup of these stressors diminishes cognition by impairing critical processes such as learning and memory. However, across the human population, cognition does not decline uniformly with age, or quantity of neurotoxic insults. Some individuals appear particularly vulnerable to stress induced cognitive impairments, while others are highly resilient. Genomic studies have identified genetic polymorphisms that correlate with accelerated cognitive decline and genetic variants that may even provide cognitive resilience. Other factors, such as lifestyle, diet, or certain prescribed drugs, also accelerate stress induced cognitive impairments. Unfortunately, our mechanistic understanding of these processes is rather limited. Much of what we know is from epidemiologic correlation studies. The vast genetic and lifestyle diversity among humans introduces numerous confounding variables that are problematic to control for, making it difficult to pinpoint causal factors and underlying mechanisms. Due to these limitations, it is nearly impossible to identify individuals during the earliest stages of cognitive degeneration and there are currently no therapeutic strategies to prevent or reverse its course. We develop induced pluripotent stem cell (iPSC)-based in vitro micro-physiological models of the human brain that incorporate CRISPR/CAS9 genome editing to precisely determine how genetic risk variants promote pathogenic phenotypes associated with cognitive decline, identify biomarkers, and develop therapeutic strategies around these insights. We recently applied our microphysiological systems (MPS) models to determine how the strongest genetic risk factor for Alzheimer's disease (AD), APOE4, causes cerebrovascular disease (Nature Med., 2020) and hypomyelination (under review at Nature). Over the past two years, we developed the multi-cellular integrated brain-on-a-chip (miBrain) as a project funded by the National Institutes of Health (NIH) National Institute of Neurological Disorders and Stroke (NINDS) (1UG3NS115064-01). The miBrain incorporates nine cell-types from the human brain (inhibitory and excitatory neurons, astrocytes, microglia, oligodendrocytes, oligodendrocyte precursor cells/OPCs, endothelial cells, pericytes, and vascular fibroblasts). Cells within the miBrain organize to form physiological tissues of the human brain, including the blood-brain barrier, neuronal circuits, myelinated axons, and neuroinflammation. We iteratively optimized and validated the miBrain via histological and transcriptomic comparison to the post-mortem human brain. We validated the miBrain can investigate disease states such as those induced by the SARs-COV2 virus and genetic AD risk factors. The majority of our studies have focused on acute neurological responses. Our preliminary studies have determined it can be cultured for at least 6 months. We are eager to apply and validate the miBrain platform further for long-term longitudinal studies. Here, we propose to apply the miBrain to longitudinally investigate neurological and cerebrovascular responses to common neurotoxic stresses commonly encountered on Earth and during spaceflight. Our central hypothesis is that genetics and drugs interact with neurotoxic stress to influence the physiological and pathological response of human brain tissue. In Aim 1, we will optimize the miBrain platform long-term longitudinal studies, integrating an automated feeding system and low cost mini-microscopes for continuous non-invasive monitoring. In Aim 2, we will construct and validate a panel of multiplexed assays for continuously monitoring miBrain response to acute and chronic exposure neurotoxic stress. In Aim 3, we will pinpoint how genetics interact with neurotoxic stress and contribute to the functions and dysfunctions of the human brain. We will also deploy the miBrain-screening platform to identify the long-term effects of FDA-approved drugs on the pathophysiological response to neurotoxic stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study will yield a platform for in vitro experimentation that can inform the design of clinical trials, enabling them to be stratified based on genetic factors. This approach can also be combined with additional variables such as spaceflight to investigate multivariate interactions between genetics, drugs, and stressors. This study will build a foundation of knowledge and technology for determining an individual's probability of adverse neurological events in space and on Earth.
Task Progress:	No reporting received. Please see the Cumulative Bibliography for published research supported by this Inter-Agency contract (Ed., 7/15/26).
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Other Journals or Periodicals	Mesentier-Louro LA, Goldman C, Ndayisaba A, Buonfiglioli A, Rooklin RB, Schuldt BR, Uchitelev A, Khurana V, Blanchard JW. "Cholesterol-mediated lysosomal dysfunction in APOE4 astrocytes promotes α-synuclein pathology in human brain tissue." bioRxiv preprint server. Posted February 14, 2025. https://doi.org/10.1101/2025.02.09.637107 , Feb-2025
Articles in Other Journals or Periodicals	Goldman C, Kareva T, Sarrafha L, Schuldt BR, Sahasrabudhe A, Ahfeldt T, Blanchard JW. "Genetically Encoded and Modular SubCellular Organelle Probes (GEM-SCOPe) reveal lysosomal and mitochondrial dysfunction driven by PRKN knockout." bioRxiv preprint server. Posted June 2, 2025. https://doi.org/10.1101/2024.05.21.594886 , Jun-2025
Articles in Other Journals or Periodicals	Joly J, Budamagunta V, Zhang Z, Nortman B, Jouzi M, Bhatnagar R, Egertson JD, Flaster ME, Grothe R, Guha S, Kaneshige K. "Large-scale single-molecule analysis of tau proteoforms." bioRxiv preprint server. Posted June 27, 2025. https://doi.org/10.1101/2025.06.26.660445 , Jun-2025
Articles in Other Journals or Periodicals	Schuldt BR, Haworth-Staines D, Perez-Arevalo A, Broekaart DW, Wang L, Gallagher G, Rabinowitz G, Goate AM, Raj T, Pereira AC, Blanchard JW. "POE4 promotes cerebrovascular fibrosis and amyloid deposition via a pericyte-to-myofibroblast transition." bioRxiv preprint server. Posted Sep 9, 2025. https://doi.org/10.1101/2025.09.04.674192 , Sep-2025
Articles in Peer-reviewed Journals	Goldman C, Kareva T, Sarrafha L, Schuldt BR, Sahasrabudhe A, Ahfeldt T, Blanchard JW. "Genetically encoded and modular subcellular organelle probes reveal dysfunction in lysosomes and mitochondria driven by PRKN knockout." iScience. 2025 July 18;28(7):112816. http://dx.doi.org/10.1016/j.isci.2025.112816 ; PMID: 40612515; PMCID: PMC12221716 , Jul-2025